# MSG No. 18.22 $P_B2_12_12$ [ Type IV, orthorhombic ]

Table 1: Wyckoff site: `4a`, site symmetry: `..2`

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\ 0,\ z]$ | `[1,4]` |
| 2 | $\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$ | `[2,3]` |
| 3 | $\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$ | `[5,8]` |
| 4 | $\left[0,\ \frac{1}{2},\ \frac{1}{2}-z\right]$ | `[6,7]` |

Table 2: Wyckoff site: `4b`, site symmetry: `2'..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\ \frac{1}{4},\ \frac{3}{4}\right]$ | `[1,6]` |
| 2 | $\left[x+\frac{1}{2},\ \frac{1}{4},\ \frac{1}{4}\right]$ | `[2,5]` |
| 3 | $\left[\frac{1}{2}-x,\ \frac{3}{4},\ \frac{1}{4}\right]$ | `[3,8]` |
| 4 | $\left[-x,\ \frac{3}{4},\ \frac{3}{4}\right]$ | `[4,7]` |

Table 3: Wyckoff site: `8c`, site symmetry: `1`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ y,\ z]$ | `[1]` |
| 2 | $\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$ | `[2]` |
| 3 | $\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ -z\right]$ | `[3]` |
| 4 | $[-x,\ -y,\ z]$ | `[4]` |
| 5 | $\left[x+\frac{1}{2},\ y,\ z+\frac{1}{2}\right]$ | `[5]` |
| 6 | $\left[x,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$ | `[6]` |
| 7 | $\left[-x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$ | `[7]` |
| 8 | $\left[\frac{1}{2}-x,\ -y,\ z+\frac{1}{2}\right]$ | `[8]` |